NSTI INNOVATION SHOWCASE | TEAM ODYSSEY

AUH Twin

Connecting aviation data to hotel demand

Monthly international hotel-demand estimates across 45 nationality markets.

Can aviation data help estimate international hotel demand in Abu Dhabi?

Tourism planners • Aviation teams • Hotel strategists

18.4%

error by nationality and month, up to 7 months ahead

WMAPE, average over 23 start dates. Repeating last year: 26.7%.

17.07%

guest-day MAPE, 1 month ahead

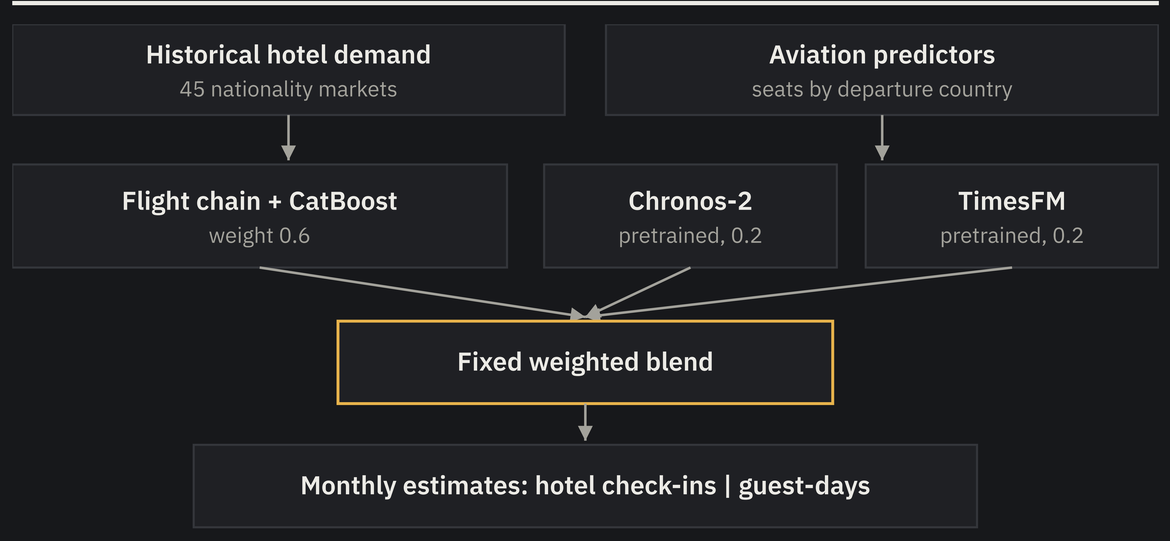
Apr 2024 to Jul 2025, 16 months. Repeating last year: 23.68%.

7.3%

error on the monthly total

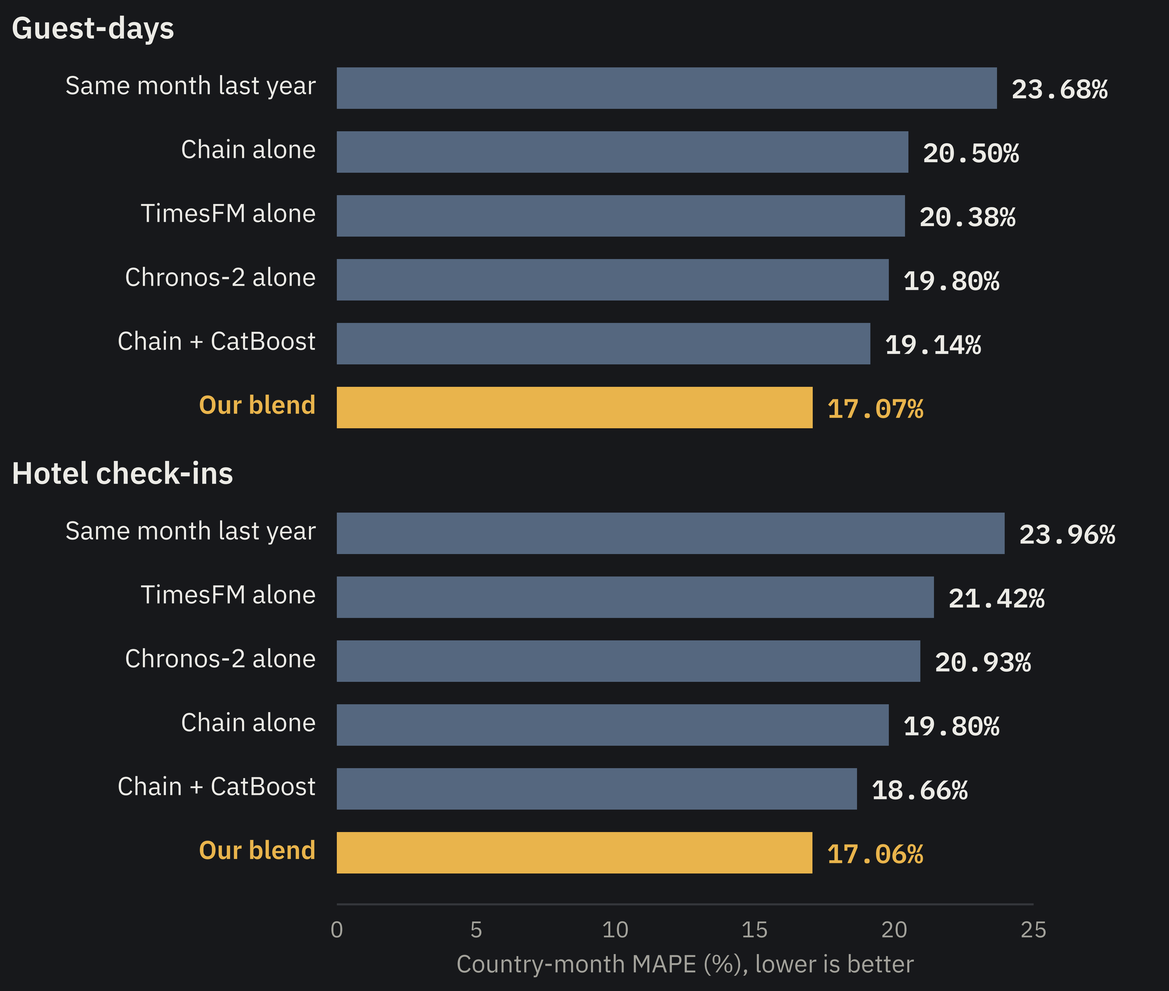
Same 23 start dates. Repeating last year: 21.3%.

FIG. 1 HOW THE MODEL WORKS



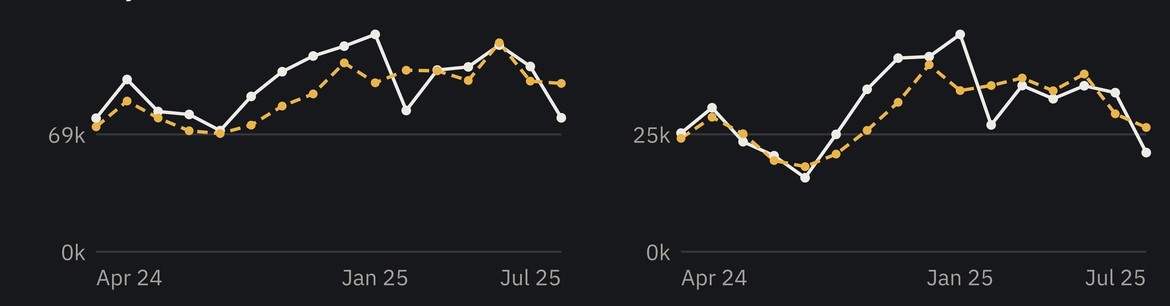
Departure country is a nationality proxy. Chronos-2 and TimesFM are pretrained, not trained by us.

FIG. 2 MODEL COMPARISON, 1 MONTH AHEAD



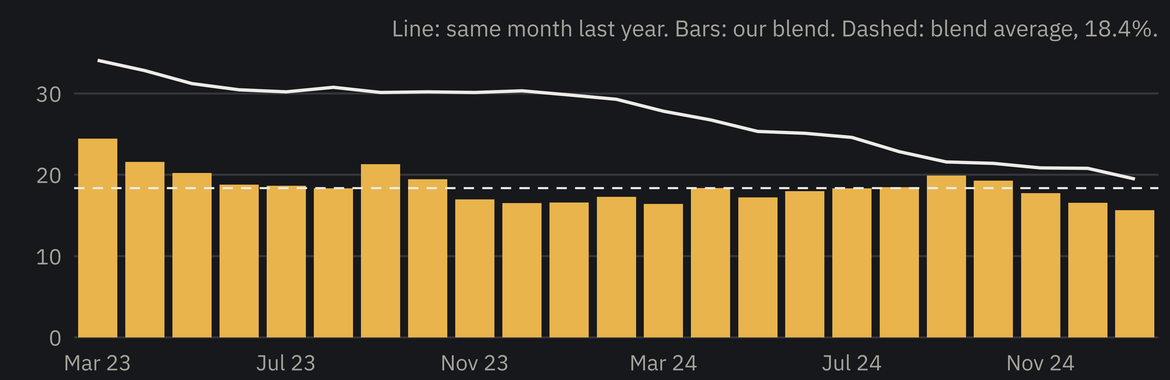
Same test for all: Apr 2024 to Jul 2025, 45 markets, complete months only (717 guest-day and 702 check-in labels of 720), using the month's filed seats.

FIG. 3 ACTUAL AGAINST ESTIMATED, LARGEST MARKET



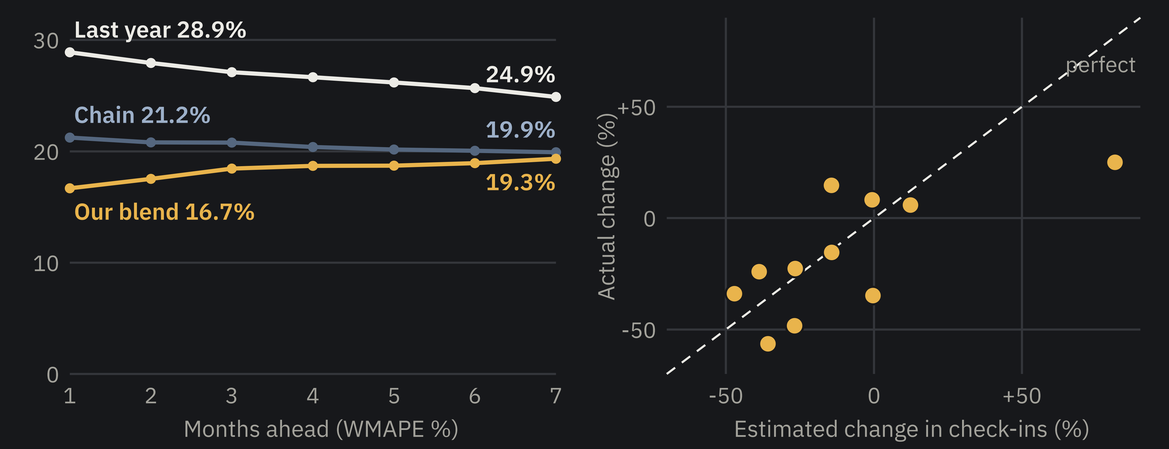
Solid: actual. Dashed: our blend, 1 month ahead. India chosen as the largest market by guest-days. An illustrative view.

FIG. 4 ERROR AT EVERY START DATE



Same recipe, 23 start dates, guest-days. It beats last year at all 23, and the chain alone at 16. Windows overlap, so these are not independent tests.

FIG. 5 HOW FAR AHEAD, AND DO FLIGHT CHANGES RESPOND?



Left: error by forecast distance, 23 start dates. Right: 11 real airline entries and exits against unchanged markets. Actual moves ran at about 0.6 to 0.8 of the estimate. Checked after a fix to weak routes, so not independent proof.

IMPACT FOR PLANNERS

+115,000
guest-days a year from one more daily London flight

+507,000
a year from the best mix of 5,000 extra weekly seats (+388,000 in a bad year)

2.4M
a year at risk if all India flights stop

HOW IT WAS TESTED

- Each forecast uses only earlier data.
- One fixed recipe at all 23 dates, real DCT data only.
- Every score can be rebuilt from saved code.
- Code reviewed by two independent AI reviewers.

LIMITATIONS

- Test months were seen during development, so scores may be optimistic.
- Departure country is only a proxy for nationality.
- Chronos-2 and TimesFM were released after some test months.
- Guest-days are not confirmed as guest nights.
- Impact figures are model estimates, not causal effects.

NEXT STEP

Evaluate on untouched data or run prospectively.

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Try it at our booth, or scan
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